

Medical Insurance Cost – Statistical Analysis & Interactive Dashboard

Lab-4: Applied Statistical Modeling & Interactive Web Dashboard

Course: Statistical Modeling with Python

Course Code: DS602

Program: M.Sc. Data Science

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1. Executive Summary

This project performs a complete statistical analysis of medical insurance charges using exploratory data analysis, descriptive statistics, hypothesis testing, multiple linear regression, multicollinearity analysis, regression diagnostics, and prediction. A separate Streamlit application provides interactive exploration and prediction.

The dataset contains 1,338 observations and 7 variables. No missing values were detected. One duplicate row was identified during the notebook data-quality check. Mean insurance charges are ₹13,270.42, while the median is ₹9,382.03.

Smokers and non-smokers show a statistically significant difference in charges using the Mann-Whitney U test ($p = 5.2702e-130$). The Chi-Square test of smoker status versus region was not significant ($p = 0.0617$).

The OLS model gives $R^2 = 0.7509$ and adjusted $R^2 = 0.7494$. The overall F-test is significant ($F = 500.8107$, $p = 0.0000e+00$). Diagnostics indicate non-normal residuals and heteroscedasticity.

Live application link: [Interactive Streamlit Dashobard](#)

2. Dataset and Data Quality

Variable	Type	Description
age	Numerical	Age of the primary beneficiary
sex	Categorical	Sex of the beneficiary
bmi	Numerical	Body Mass Index
children	Numerical	Number of children/dependents
smoker	Categorical	Smoking status
region	Categorical	Residential region
charges	Numerical	Medical insurance charges; regression target

The notebook checks dataset shape, column names, data types, missing values, and duplicate rows before analysis. The `analysis.py` module standardizes numerical and categorical fields and handles required missing values.

3. Descriptive Statistics

Feature	Mean	Median	Std Dev	IQR	Skewness	Kurtosis
age	39.2070	39.0000	14.0500	24.0000	0.0557	-1.2451
bmi	30.6634	30.4000	6.0982	8.3975	0.2840	-0.0507
children	1.0949	1.0000	1.2055	2.0000	0.9384	0.2025
charges	13270.4223	9382.0330	12110.0112	11899.6254	1.5159	1.6063

Charges are positively skewed (skewness = 1.5159), indicating a long right tail. Age is comparatively symmetric, while children is positively skewed because it is a discrete count variable.

4. Distribution Analysis

The complete individual distribution graphs are included below. Each variable has its own graph so that no portion of a graph is cut off.

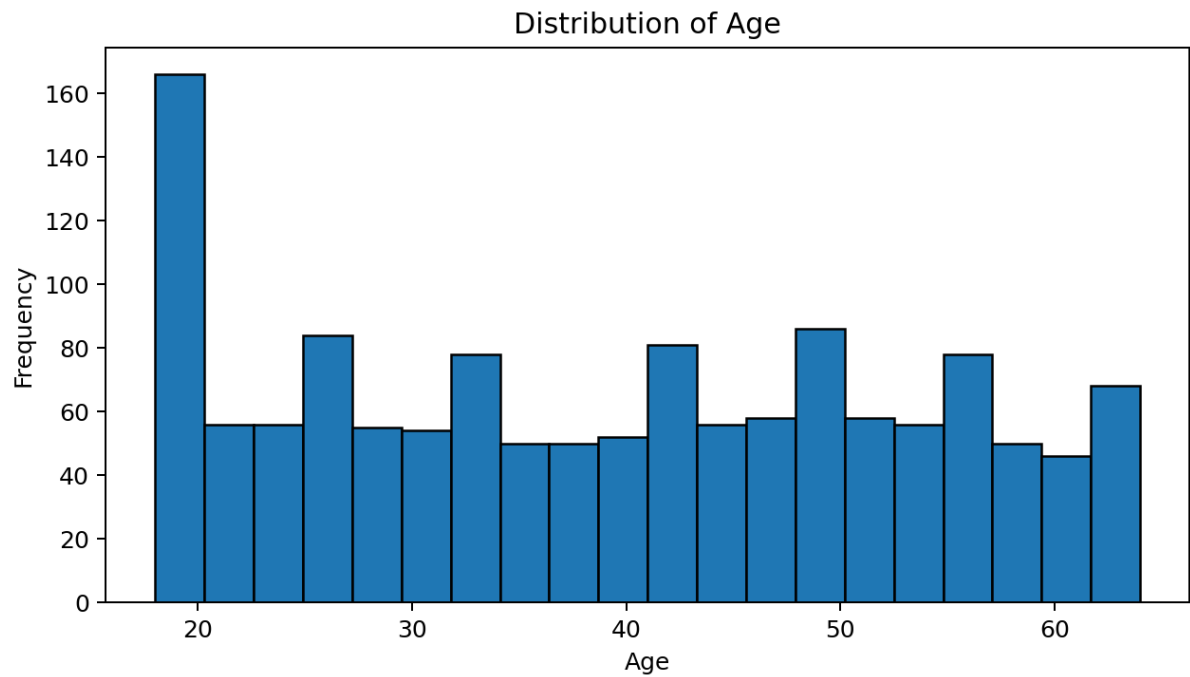


Figure: Distribution of Age

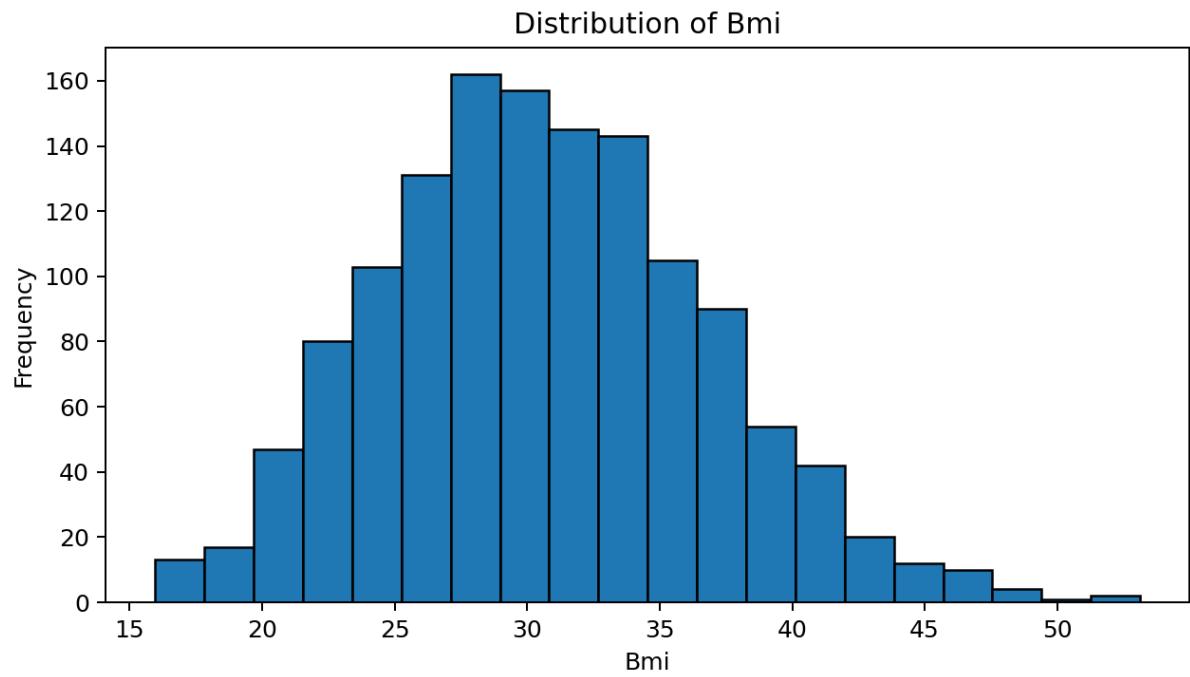


Figure: Distribution of Bmi

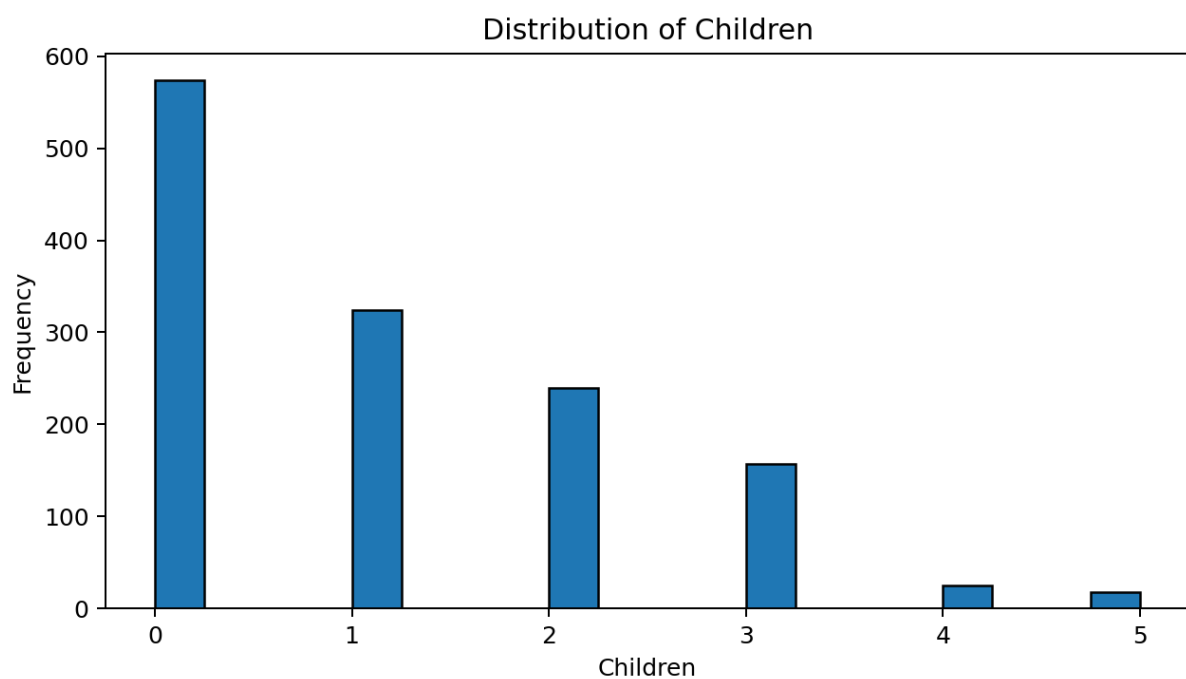


Figure: Distribution of Children

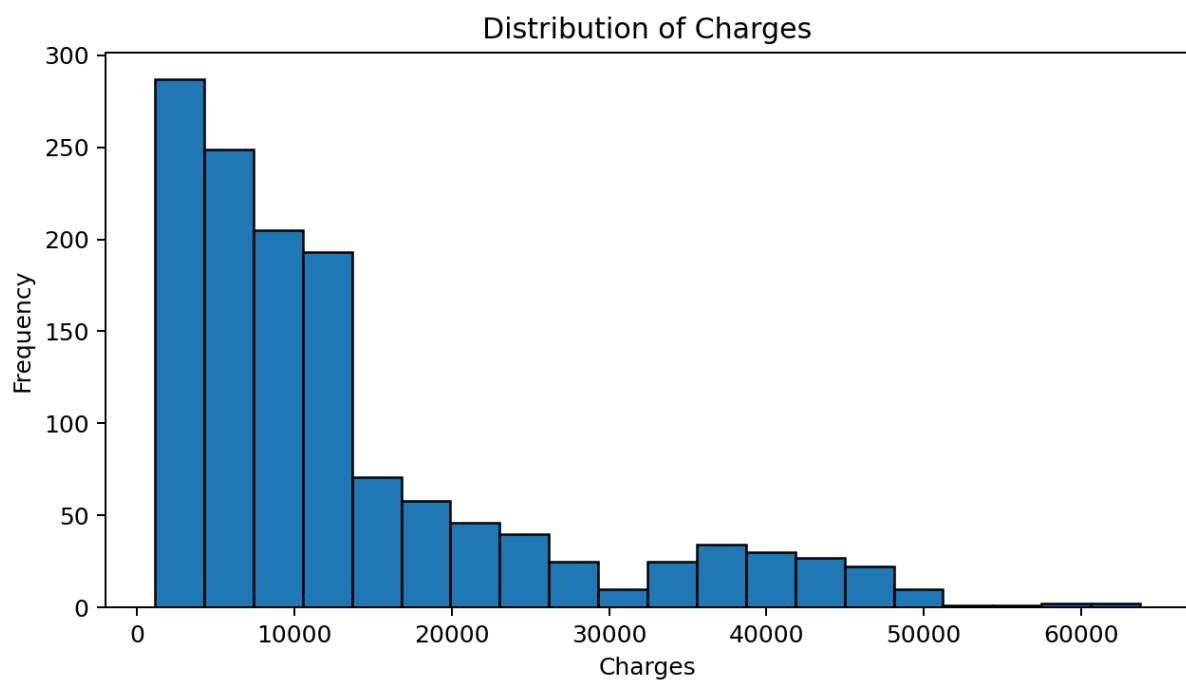


Figure: Distribution of Charges

5. Boxplot and Outlier Analysis

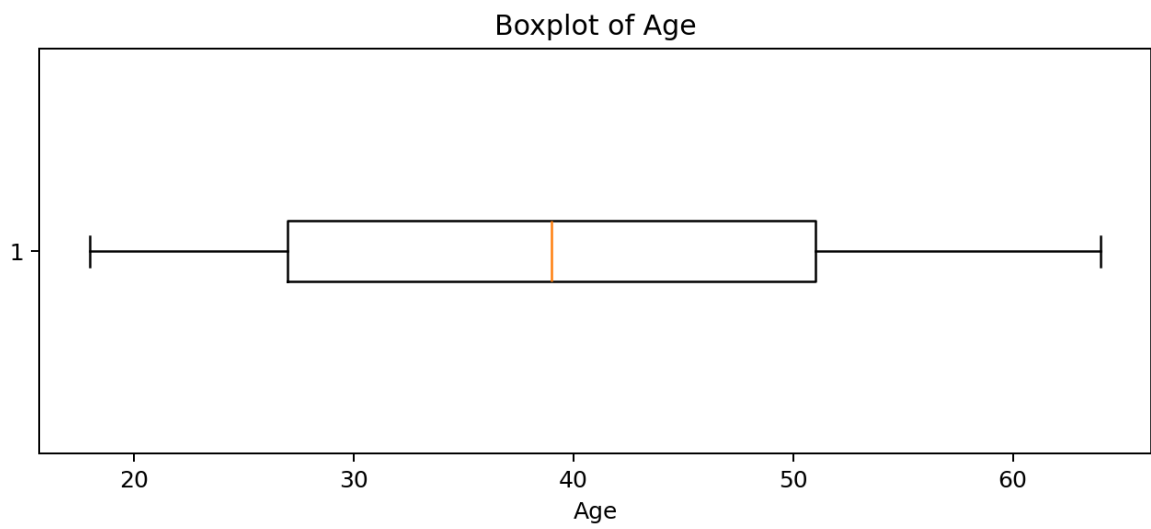


Figure: Boxplot of Age

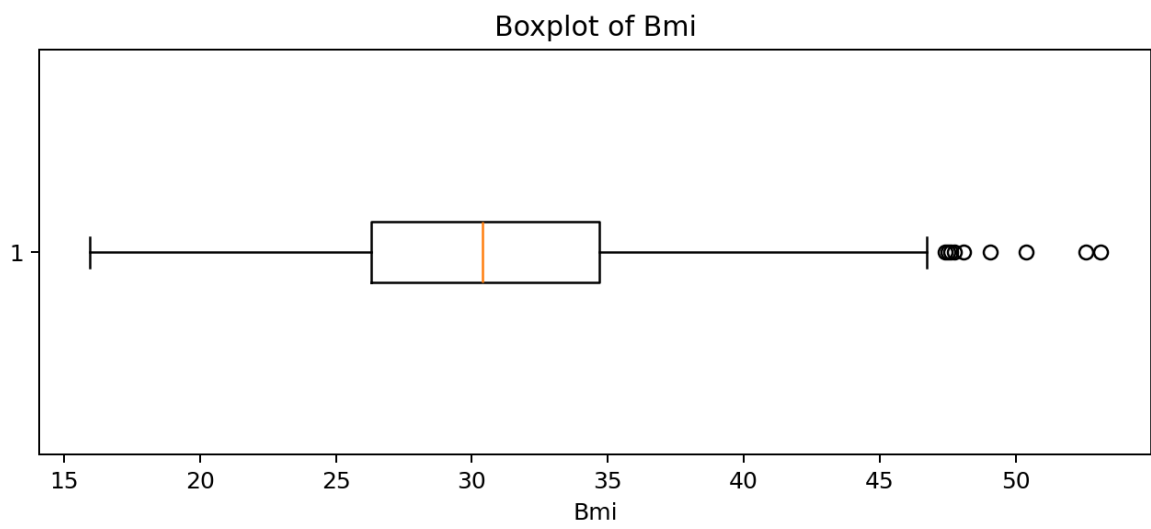


Figure: Boxplot of Bmi

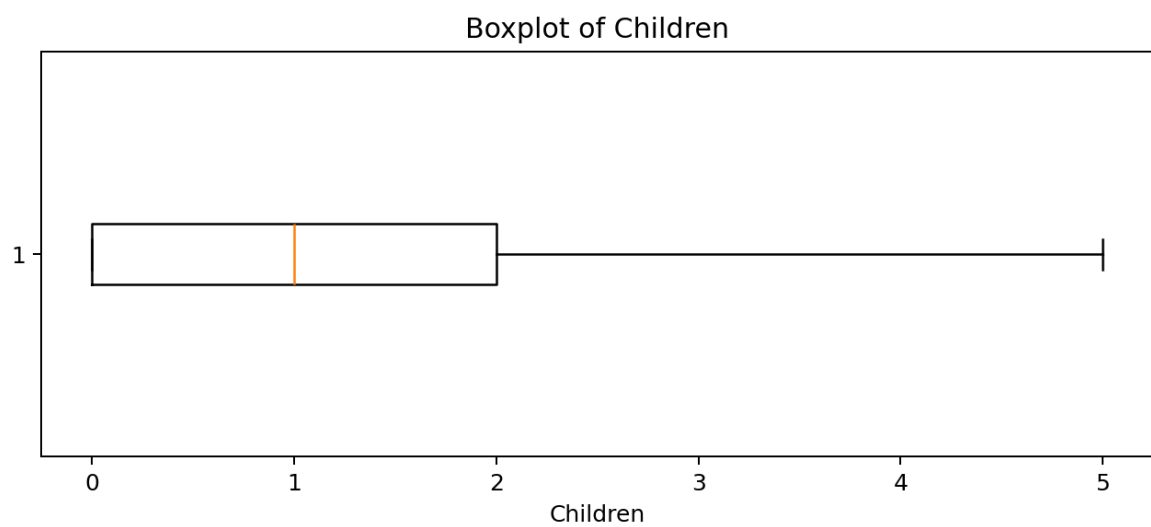


Figure: Boxplot of Children

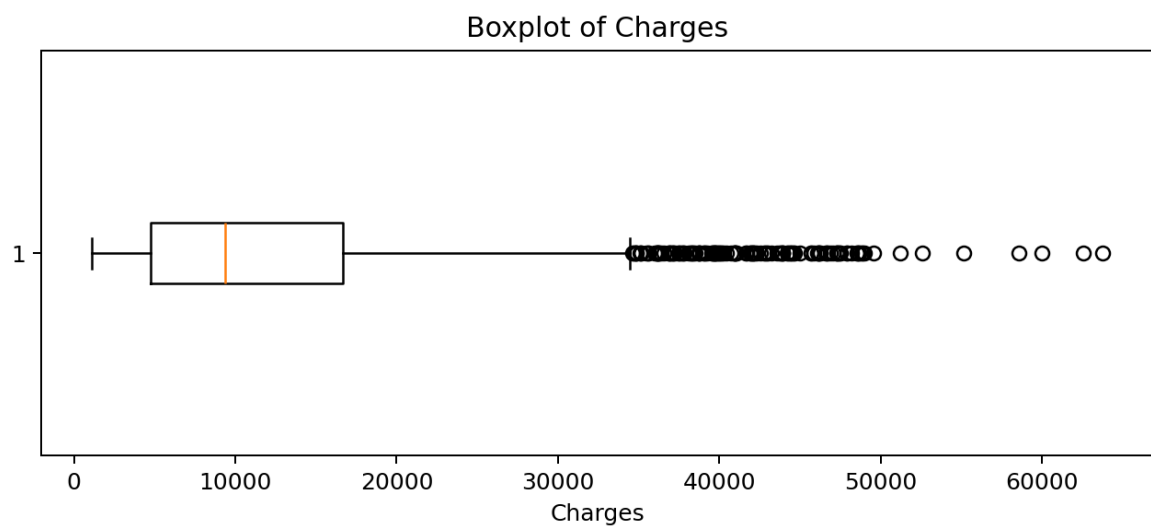
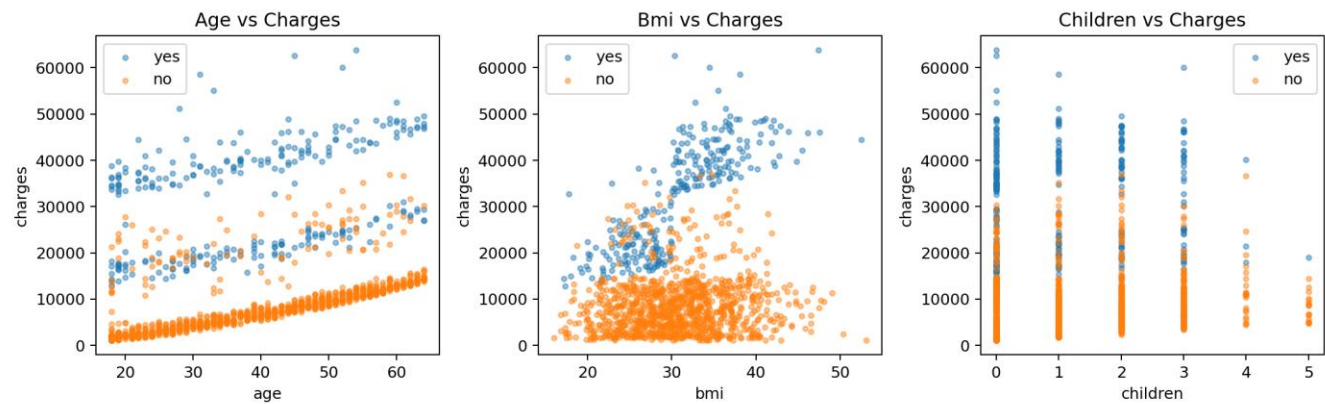


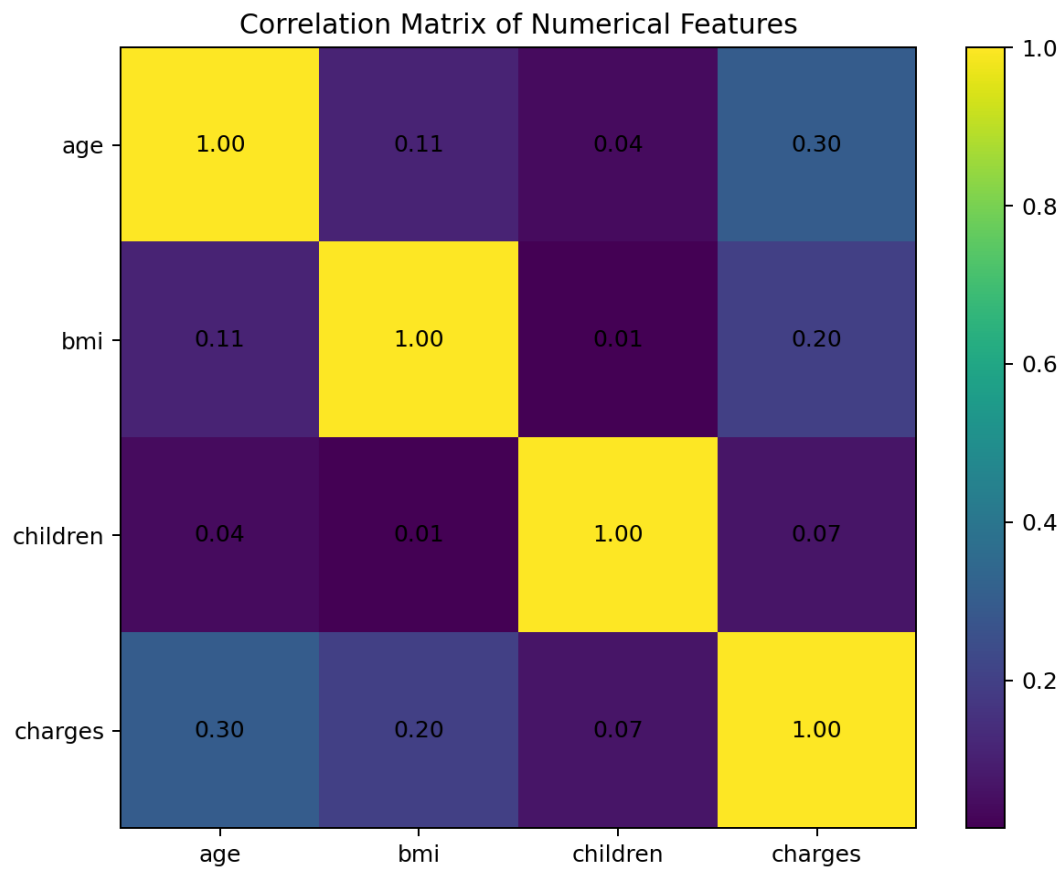
Figure: Boxplot of Charges

The charges boxplot shows many high-value observations beyond the upper whisker. These observations were retained because they can represent genuine medical-cost variation and are relevant to the prediction problem.

6. Bivariate Analysis and Correlation



The scatter plots show relationships between age, BMI, children, and insurance charges, with smoker status used to distinguish observations.



Pearson correlations with charges are: age = 0.2990, BMI = 0.1983, and children = 0.0680.

7. Hypothesis Testing

7.1 Hypothesis Test 1 – Two-Group Comparison

Grouping variable: smoker. Numerical metric: charges.

H0: There is no statistically significant difference between the two groups.

H1: There is a statistically significant difference between the two groups.

Test / Measure	Result	Decision
Smoker	n = 274, mean = ₹32,050.23	—
Non-smoker	n = 1064, mean = ₹8,434.27	—
Shapiro-Wilk – smoker	W = 0.939552, p = 3.6250e-09	Reject normality
Shapiro-Wilk – non-smoker	W = 0.872864, p = 1.4459e-28	Reject normality
Levene	Statistic = 332.6135, p = 1.5593e-66	Reject equal variance
Mann-Whitney U	U = 284133.0, p = 5.2702e-130	Reject H0

Because both groups failed the normality check, the required decision path selects the Mann-Whitney U test. The very small p-value provides strong evidence that charge distributions differ between smokers and non-smokers.

7.2 Hypothesis Test 2 – Chi-Square Test of Independence

H0: Smoker status and region are independent.

H1: Smoker status and region are associated.

Smoker	northeast	northwest	southeast	southwest
no	257	267	273	267
yes	67	58	91	58

Chi-Square statistic = 7.3435, df = 3, p-value = 0.0617. Since $p \geq 0.05$, H0 is not rejected. There is no statistically significant evidence of an association between smoker status and region at the 5% significance level.

8. Multiple Linear Regression – OLS

The target variable is charges. Predictors are age, BMI, children, sex, smoker, and region. Categorical variables are dummy encoded using one reference category.

Model fit: $R^2 = 0.7509$; Adjusted $R^2 = 0.7494$; F-statistic = 500.8107; overall F-test p-value = 0.0000e+00.

Term	Coefficient	p-value	95% CI Lower	95% CI Upper
const	-11938.5386	5.5790e-32	-13876.3934	-10000.6837
age	256.8564	7.7832e-89	233.5138	280.1989
bmi	339.1935	6.4982e-31	283.0884	395.2985
children	475.5005	5.7697e-04	205.1633	745.8378
sex_male	-131.3144	6.9335e-01	-784.4703	521.8416
smoker_yes	23848.5345	0.0000e+00	23038.0307	24659.0384
region_northwest	-352.9639	4.5877e-01	-1287.2982	581.3704
region_southeast	-1035.0220	3.0782e-02	-1974.0968	-95.9473
region_southwest	-960.0510	4.4765e-02	-1897.6364	-22.4656

At the 5% level, age, BMI, children, smoker_yes, region_southeast, and region_southwest are statistically significant in this fitted model. Sex_male and region_northwest are not significant.

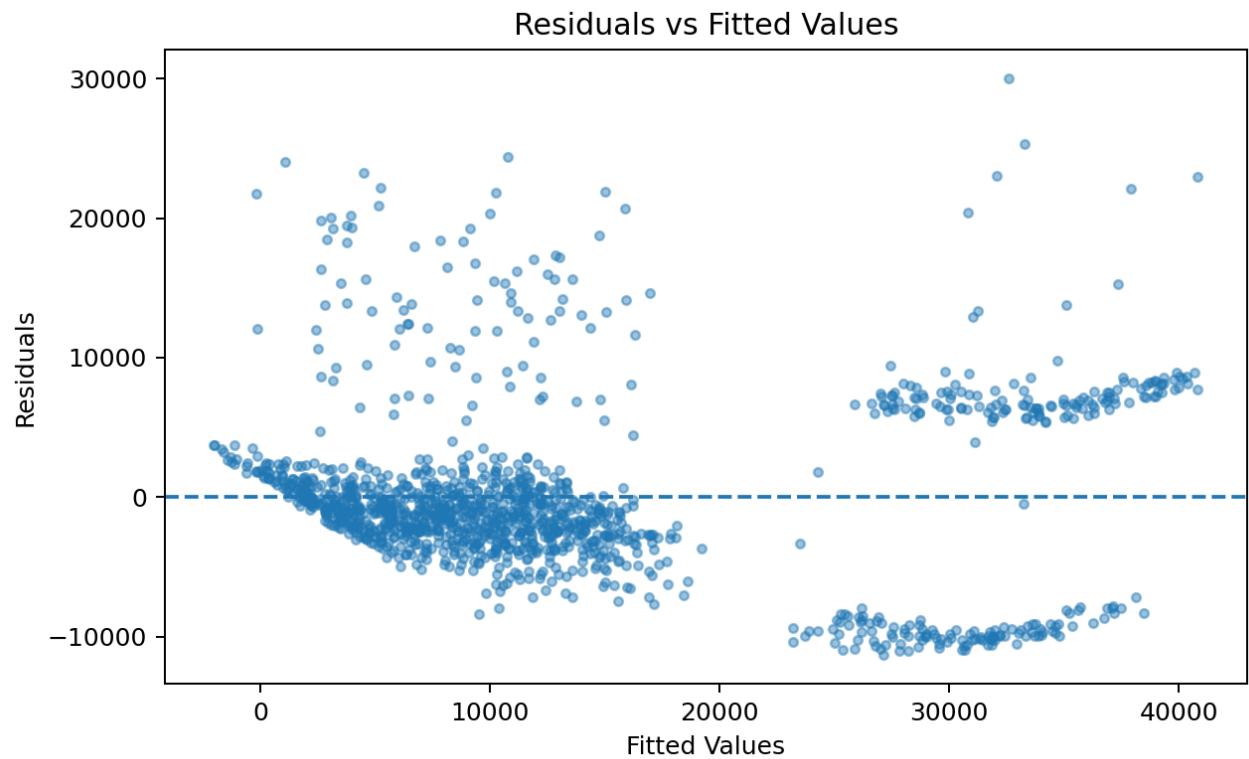
9. Multicollinearity and Regression Diagnostics

9.1 Variance Inflation Factor (VIF)

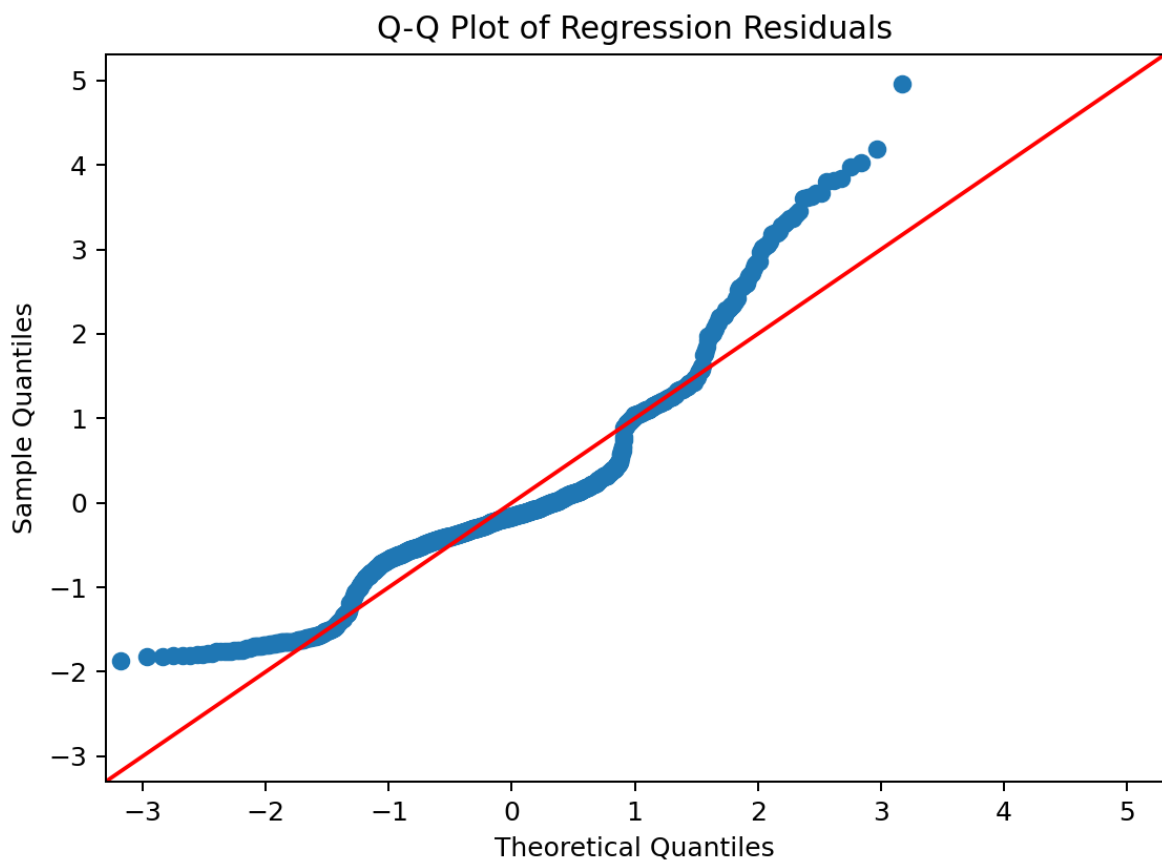
Continuous Predictor	VIF
age	1.0138
bmi	1.0122
children	1.0019

All continuous-predictor VIF values are close to 1, indicating no serious multicollinearity among age, BMI, and children.

9.2 Residual Diagnostics



The residuals-vs-fitted plot is used to inspect linearity and constant variance.



Jarque-Bera statistic = 718.8873, $p = 7.8635e-157$; residual normality is rejected. Breusch-Pagan LM statistic = 121.7436, $p = 1.4467e-22$; there is strong evidence of heteroscedasticity.

10. Live Prediction

Example input used in the notebook: age = 40, BMI = 30, children = 2, sex = male, smoker = no, region = southeast.

Prediction Output	Value
Predicted charge	₹9,462.52
95% prediction interval – lower	₹-2,456.56
95% prediction interval – upper	₹21,381.60

The prediction interval is wide because individual medical costs have substantial variability.

11. Streamlit Interactive Dashboard

The Streamlit dashboard is maintained separately from the detailed notebook analysis. It provides interactive data exploration, hypothesis testing, live prediction, and diagnostic visualizations. The dashboard code was not changed as part of the notebook/report preparation.

[Interactive Streamlit Dashboard](#)

12. Overall Findings and Conclusion

- Data quality: 1,338 observations and 7 variables were analyzed; no missing values were detected and one duplicate row was identified.
- Charges: mean ₹13,270.42, median ₹9,382.03, with positive skewness of 1.5159.
- Smoking effect: smokers had a much higher average charge (₹32,050.23) than non-smokers (₹8,434.27); Mann-Whitney $p = 5.2702e-130$.
- Smoker-region association: Chi-Square $p = 0.0617$; no significant association was detected at $\alpha = 0.05$.
- Regression fit: $R^2 = 0.7509$, so approximately 75.1% of sample variation in charges is explained by the fitted predictors.
- Multicollinearity: VIF values around 1 indicate no serious multicollinearity among the continuous predictors.
- Diagnostics: residual normality and constant variance assumptions are not fully satisfied, so OLS inference should be interpreted with these limitations in mind.

Overall, smoking status is the strongest practical signal in this dataset, while the multiple linear regression model provides substantial explanatory power. The analysis should be interpreted as an empirical statistical model and not as proof of causation.

Appendix – Technical Components

The project uses Pandas and NumPy for data handling, SciPy for hypothesis tests, Statsmodels for OLS regression and diagnostics, Seaborn/Matplotlib for statistical visualizations, and Streamlit/Plotly for the interactive dashboard. The notebook reuses the analytical functions implemented in `analysis.py`.

Regression specification:

$$\text{charges} = \beta_0 + \beta_1(\text{age}) + \beta_2(\text{bmi}) + \beta_3(\text{children}) + \beta_4(\text{sex}) + \beta_5(\text{smoker}) + \beta_6(\text{region}) + \varepsilon$$

Decision rule: $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$; $p\text{-value} \geq 0.05 \rightarrow \text{Fail to reject } H_0$.

Project deliverables: analysis.py, analysis.ipynb, app.py, data/insurance.csv, requirements.txt, README.md, and this final report.